PBL Portal Content - Deepak Kumar

Smart Hospital Management System (SmartCare) | Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri

Member	Deepak Kumar (24E1ARADM30P035)
Module	Module 2 - Patient and Doctor Management
Code owned	Patient, Doctor, Receptionist models, PatientDAO, DoctorDAO, ReceptionistDAO, PatientServlet, DoctorServlet, patient JSPs (dashboard, profile, medical records), doctor JSPs (dashboard, patient profile, records), receptionist patient registration.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

How to use this file

1) Daily Work Log tab: pick the Date, type Hours Worked, keep Status as shown, paste the description, click Submit Today's Work. 2) Weekly Report tab: choose the Week, paste the GitHub link, then paste the three boxes (Work Completed, Problems Faced, Plan for Next Week) from Part A. 3) Part A (weekly) comes first, Part B has the day-wise table. Submit your own entries only, from your own login.

PART A - Weekly Reports (13 weeks)

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Joined team of five and discussed hospital problems the project could solve. (2) Collected how patients and doctors are registered in small hospitals. (3) Wrote the abstract section on patient and doctor management. (4) Discussed the tech stack with the guide and the team. (5) Reviewed the final abstract and helped submit it.
Problems Faced	Team had many different ideas. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed patient requirements: registration, profile, history and records. (2) Listed doctor requirements: profile, department, specialisation and availability. (3) Wrote data requirements for patient and doctor details.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Listed patient requirements: registration, profile, history and records. (2) Listed doctor requirements: profile, department, specialisation and availability. (3) Wrote data requirements for patient and doctor details. (4) Added constraints such as unique patient ID and mandatory fields. (5) Reviewed the patient and doctor sections of the SRS with the team.
Problems Faced	Overlap between patient and receptionist features. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for Patient and Doctor actors. (2) Designed the class diagram for Patient with attributes and methods. (3) Designed the class diagram for Doctor and Receptionist.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Drew the use-case diagram for Patient and Doctor actors. (2) Designed the class diagram for Patient with attributes and methods. (3) Designed the class diagram for Doctor and Receptionist. (4) Drew the sequence diagram for registering a new patient. (5) Combined the module UML with the team diagrams.
Problems Faced	Deciding attributes for Patient class. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the patients table with personal and contact fields. (2) Designed the doctors table with department link and qualification. (3) Designed the medical_records table and its relation to patients.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Designed the patients table with personal and contact fields. (2) Designed the doctors table with department link and qualification. (3) Designed the medical_records table and its relation to patients. (4) Set up local environment (JDK 17, Tomcat, MySQL) and cloned the repo. (5) Drew patient registration and doctor dashboard mock-ups.
Problems Faced	Normalising patient and record tables. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Patient model class with all fields. (2) Wrote Doctor and Receptionist model classes. (3) Implemented PatientDAO insert and select methods.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote Patient model class with all fields. (2) Wrote Doctor and Receptionist model classes. (3) Implemented PatientDAO insert and select methods. (4) Added PatientDAO update, search and list methods. (5) Implemented DoctorDAO with department-wise doctor listing.
Problems Faced	Date conversion between java.util and java.sql. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Implemented PatientServlet actions for view and update profile. (2) Implemented DoctorServlet dashboard action with today's data. (3) Added the patient search action by name and phone.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Implemented PatientServlet actions for view and update profile. (2) Implemented DoctorServlet dashboard action with today's data. (3) Added the patient search action by name and phone. (4) Implemented ReceptionistDAO and the registration action. (5) Tested servlet actions with sample data.
Problems Faced	Null pointer when patient had no record. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the patient dashboard.jsp with quick links. (2) Created the patient profile.jsp with editable form. (3) Created the doctor dashboard.jsp with appointment summary.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Created the patient dashboard.jsp with quick links. (2) Created the patient profile.jsp with editable form. (3) Created the doctor dashboard.jsp with appointment summary. (4) Created the doctor patient_profile.jsp showing history. (5) Created the receptionist register_patient.jsp and patients list.
Problems Faced	Form values lost after validation failure. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added server-side validation for phone, age and blood group. (2) Added JavaScript validation on registration form fields. (3) Handled exceptions in DAO methods with clear user messages.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Added server-side validation for phone, age and blood group. (2) Added JavaScript validation on registration form fields. (3) Handled exceptions in DAO methods with clear user messages. (4) Used JSTL and escaped output to avoid script injection. (5) Added pagination for the patient list.
Problems Faced	Search became slow without proper query. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed the medical records view for the patient. (2) Added the doctor's records.jsp to view and add notes. (3) Added patient auto-ID generation.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Completed the medical records view for the patient. (2) Added the doctor's records.jsp to view and add notes. (3) Added patient auto-ID generation. (4) Reviewed code, cleaned names and added comments. (5) Pushed module code to GitHub and updated the sheet.
Problems Faced	Time was short for comments and cleanup. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Integrated patient registration with the login of the patient user. (2) Linked doctor listing with the appointment booking page. (3) Checked navigation between patient and doctor pages after merge.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Integrated patient registration with the login of the patient user. (2) Linked doctor listing with the appointment booking page. (3) Checked navigation between patient and doctor pages after merge. (4) Fixed DAO connection leaks found during integration. (5) Ran an end-to-end flow from registration to profile view.
Problems Faced	Connections were not closed in a few methods. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for patient registration and profile update. (2) Tested invalid and boundary inputs on forms. (3) Tested doctor dashboard data with multiple doctors.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote test cases for patient registration and profile update. (2) Tested invalid and boundary inputs on forms. (3) Tested doctor dashboard data with multiple doctors. (4) Tested search and pagination on patient list. (5) Fixed defects found and retested.
Problems Faced	Some inputs allowed negative age. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked patient and doctor pages on the deployed site. (2) Fixed date format issue on the cloud database. (3) Added missing seed data for departments and doctors.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Checked patient and doctor pages on the deployed site. (2) Fixed date format issue on the cloud database. (3) Added missing seed data for departments and doctors. (4) Retested registration on the live server. (5) Fixed layout issues on smaller screens.
Problems Faced	Date format differed on cloud MySQL. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the patient and doctor module section of the report. (2) Added screenshots and table descriptions to the report. (3) Prepared module slides for the presentation.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	(1) Wrote the patient and doctor module section of the report. (2) Added screenshots and table descriptions to the report. (3) Prepared module slides for the presentation. (4) Practised the demo of patient registration and doctor view. (5) Prepared viva answers on DAO and JDBC.
Problems Faced	Report formatting took extra time. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

PART B - Daily Work Log (Monday to Friday, 65 entries)

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description
06-07-2026 (Mon)	2	Completed	Joined team of five and discussed hospital problems the project could solve.
07-07-2026 (Tue)	1.75	Completed	Collected how patients and doctors are registered in small hospitals.
08-07-2026 (Wed)	1.5	Completed	Wrote the abstract section on patient and doctor management.
09-07-2026 (Thu)	2	Completed	Discussed the tech stack with the guide and the team.
10-07-2026 (Fri)	1.75	Completed	Reviewed the final abstract and helped submit it.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description
13-07-2026 (Mon)	1.5	Completed	Listed patient requirements: registration, profile, history and records.
14-07-2026 (Tue)	1.75	Completed	Listed doctor requirements: profile, department, specialisation and availability.
15-07-2026 (Wed)	2	Completed	Wrote data requirements for patient and doctor details.
16-07-2026 (Thu)	2	Completed	Added constraints such as unique patient ID and mandatory fields.
17-07-2026 (Fri)	1.75	Completed	Reviewed the patient and doctor sections of the SRS with the team.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description
20-07-2026 (Mon)	1.75	Completed	Drew the use-case diagram for Patient and Doctor actors.
21-07-2026 (Tue)	2	Completed	Designed the class diagram for Patient with attributes and methods.
22-07-2026 (Wed)	2	Completed	Designed the class diagram for Doctor and Receptionist.
23-07-2026 (Thu)	1.75	Completed	Drew the sequence diagram for registering a new patient.
24-07-2026 (Fri)	1.5	Completed	Combined the module UML with the team diagrams.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description
27-07-2026 (Mon)	1.75	Completed	Designed the patients table with personal and contact fields.
28-07-2026 (Tue)	1.5	Completed	Designed the doctors table with department link and qualification.
29-07-2026 (Wed)	1.75	Completed	Designed the medical_records table and its relation to patients.
30-07-2026 (Thu)	2	Completed	Set up local environment (JDK 17, Tomcat, MySQL) and cloned the repo.
31-07-2026 (Fri)	1.5	Completed	Drew patient registration and doctor dashboard mock-ups.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description
03-08-2026 (Mon)	1.5	Completed	Wrote Patient model class with all fields.
04-08-2026 (Tue)	1.5	Completed	Wrote Doctor and Receptionist model classes.
05-08-2026 (Wed)	1.5	Completed	Implemented PatientDAO insert and select methods.
06-08-2026 (Thu)	2	Completed	Added PatientDAO update, search and list methods.
07-08-2026 (Fri)	1.5	Completed	Implemented DoctorDAO with department-wise doctor listing.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description
10-08-2026 (Mon)	1.5	Completed	Implemented PatientServlet actions for view and update profile.
11-08-2026 (Tue)	1.75	Completed	Implemented DoctorServlet dashboard action with today's data.
12-08-2026 (Wed)	1.75	Completed	Added the patient search action by name and phone.
13-08-2026 (Thu)	2	Completed	Implemented ReceptionistDAO and the registration action.
14-08-2026 (Fri)	1.5	Completed	Tested servlet actions with sample data.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description
17-08-2026 (Mon)	2	Completed	Created the patient dashboard.jsp with quick links.
18-08-2026 (Tue)	1.75	Completed	Created the patient profile.jsp with editable form.
19-08-2026 (Wed)	1.75	Completed	Created the doctor dashboard.jsp with appointment summary.
20-08-2026 (Thu)	2	Completed	Created the doctor patient_profile.jsp showing history.
21-08-2026 (Fri)	2	Completed	Created the receptionist register_patient.jsp and patients list.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description
24-08-2026 (Mon)	1.5	Completed	Added server-side validation for phone, age and blood group.
25-08-2026 (Tue)	1.5	Completed	Added JavaScript validation on registration form fields.
26-08-2026 (Wed)	2	Completed	Handled exceptions in DAO methods with clear user messages.
27-08-2026 (Thu)	2	Completed	Used JSTL and escaped output to avoid script injection.
28-08-2026 (Fri)	1.5	Completed	Added pagination for the patient list.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description
31-08-2026 (Mon)	2	Completed	Completed the medical records view for the patient.
01-09-2026 (Tue)	2	Completed	Added the doctor's records.jsp to view and add notes.
02-09-2026 (Wed)	1.5	Completed	Added patient auto-ID generation.
03-09-2026 (Thu)	1.5	Completed	Reviewed code, cleaned names and added comments.
04-09-2026 (Fri)	1.75	Completed	Pushed module code to GitHub and updated the sheet.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description
07-09-2026 (Mon)	2	Completed	Integrated patient registration with the login of the patient user.
08-09-2026 (Tue)	1.75	Completed	Linked doctor listing with the appointment booking page.
09-09-2026 (Wed)	1.5	Completed	Checked navigation between patient and doctor pages after merge.
10-09-2026 (Thu)	1.5	Completed	Fixed DAO connection leaks found during integration.
11-09-2026 (Fri)	2	Completed	Ran an end-to-end flow from registration to profile view.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description
14-09-2026 (Mon)	2	Completed	Wrote test cases for patient registration and profile update.
15-09-2026 (Tue)	1.5	Completed	Tested invalid and boundary inputs on forms.
16-09-2026 (Wed)	2	Completed	Tested doctor dashboard data with multiple doctors.
17-09-2026 (Thu)	1.75	Completed	Tested search and pagination on patient list.
18-09-2026 (Fri)	2	Completed	Fixed defects found and retested.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description
21-09-2026 (Mon)	1.5	Completed	Checked patient and doctor pages on the deployed site.
22-09-2026 (Tue)	1.75	Completed	Fixed date format issue on the cloud database.
23-09-2026 (Wed)	1.5	Completed	Added missing seed data for departments and doctors.
24-09-2026 (Thu)	1.5	Completed	Retested registration on the live server.
25-09-2026 (Fri)	1.5	Completed	Fixed layout issues on smaller screens.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description
28-09-2026 (Mon)	1.5	Completed	Wrote the patient and doctor module section of the report.
29-09-2026 (Tue)	2	Completed	Added screenshots and table descriptions to the report.
30-09-2026 (Wed)	2	Completed	Prepared module slides for the presentation.
01-10-2026 (Thu)	1.75	Completed	Practised the demo of patient registration and doctor view.
02-10-2026 (Fri)	1.5	Completed	Prepared viva answers on DAO and JDBC.

Total hours in this log: 113.25 over 65 days.